

TP406 - Instruction for Team Projects -

Goal : The purpose of this team project is to solidify the concept of ‘tensor’ and deepen their understanding of some interesting tensor-related problems and how tensors appear in various situations, and provide each student with an opportunity to communicate with team members and solve problems together.

Schedule : Our timeline for this project is as follows:

- (1) (May 12, Tue) Problem suggestion.
- (2) (May 21, Thu) 1st presentation (preliminary version, slide, lunch provided)
- (3) (May 26, Tue) Intermediate check (team by team)
- (4) (June 2, Tue) 2nd presentation (poster session, lunch provided)
- (5) (June 13, Sat) Due date for submission of team report.

Evaluation

- (1) (20%) How appropriately did the team choose their problem and how accurately did they identify the problem? (via 1st proposal presentation)
- (2) (20%) How actively did the team communicate and proceed with the problem-solving process together? (via the final report and peer-review evaluation)
- (3) (40%) To what extent did the final output achieve the originally proposed objective? (focused on Novelty, Approach, Method of execution)
- (4) (20%) The potential for future development of the project

AI-policy : The use of AI technology is basically allowed. But, in the final report you **must** concretely specify how did you use AI in the project and which part(s) done by the help of AI. It will be carefully considered how properly a team did use AI in their project!

Problems :

- (1) (Problem 1 - Normal) Determine the rank of the outer product tensor μ in 3-space (over $\mathbb{R}$ or $\mathbb{C}$). Recall that Let $V = \mathbb{R}^3$. Consider the *outer product map*

$$\mu : V \times V \rightarrow V, (\mathbf{a}, \mathbf{b}) \mapsto \mathbf{a} \times \mathbf{b}.$$

Since μ is a bilinear map, $\mu \in V^* \otimes V^* \otimes V (\cong \mathbb{R}^{27})$. Take a standard basis $\{e_1, e_2, e_3\}$ of V . Consider its dual basis $\{e_1^*, e_2^*, e_3^*\}$. Then we obtain

$$\begin{aligned} \mu &= e_2^* \otimes e_3^* \otimes e_1 - e_3^* \otimes e_2^* \otimes e_1 \\ &\quad - e_1^* \otimes e_3^* \otimes e_2 + e_3^* \otimes e_1^* \otimes e_2 \\ &\quad + e_1^* \otimes e_2^* \otimes e_3 - e_2^* \otimes e_1^* \otimes e_3. \end{aligned}$$

from the definition of outer product. Thus, we see $\mathbf{R}(\mu) \leq 6$. We know that $\mathbf{R}(\mu) \leq 5$ by Team Activity 1. It is also known that $\mathbf{R}(\mu) = 4$. Thus, $\mathbf{R}(\mu) = 4$ or 5 (one comment : If you believe the rank is 5, then you will need to show inconsistency of a system of 27 polynomial equations of degree 3 in 36 unknowns. Begin with writing down the system).

- (2) (Problem 2 - Challenging) Let $V = \text{Mat}_{3,3}(\mathbb{C})$, the vector space of 3×3 -complex matrices and $M_{\langle 3 \rangle}$ be the 3×3 -matrix multiplication tensor. Then, $M_{\langle 3 \rangle} \in V^* \otimes V^* \otimes V$. It is known that $19 \leq R(M_{\langle 3 \rangle}) \leq 23$ and $\underline{R}(M_{\langle 3 \rangle}) \leq 21$.
Can you decide $R(M_{\langle 3 \rangle})$ (or $\underline{R}(M_{\langle 3 \rangle})$) or improve bounds for this? or can you suggest a new procedure to do this? (one comment: Some rank 23 decompositions are known explicitly. One may begin with studying these results first. It will be also very good if you can find a different kind of 23 summands decomposition)
- (3) (Problem 3 - Challenging) Decide $R_S^{\mathbb{R}}(x^2y^2z^2)$, the real Waring rank of $x^2y^2z^2$. It is known that $11 \leq R_S^{\mathbb{R}}(x^2y^2z^2) \leq 13$. It will be very nice if you can improve the bound or find a different kind of Waring decomposition of $x^2y^2z^2$.
- (4) (Problem 4 - Do Your Own) Based on your knowledge and interests, try to search and find a relevant tensor-related problem as your team project. Note that you can start the project after obtaining approval of mine. Please contact me as soon as possible, once your team finds some candidate(s).
- (5) (Problem 5 - Normal) “Low rank tensor approximation” Develop a method of approximating any given tensor in $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$ (i.e. Rubik’s cube case) by rank 2 tensor decomposition. In other words, for any given tensor $T \in \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$, find a best approximate rank 2 tensor $T' \in \sigma_2^0(\text{Seg}(\mathbb{C}^3 \times \mathbb{C}^3 \times \mathbb{C}^3))$ such that the distance $\|T - T'\|$ is minimal (of course, first you need to decide which distance function you will use). Also, implement it by computer code.
- (6) (Problem 6 - Rather Easy) “Blind Source Separation & Identifiability” Study what Blind Source Separation (BSS) is and how it can be related to tensor problem, in particular to *identifiability* problem. Based on the literature and known results, make a good procedure to perform BSS using tensor in a suitable tensor space and implement it by compute code.